NPC scripting overview

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This part contains 7 notes, from `npc-library/archival-scripting-notes.md` through `server-notes/setcharprop-hash-n.md`.

Narrative NPC notes under `npc-library/`, plus `server-notes/setcharprop-hash-n.md`. Script bodies stay in the `page.json` files. Image names are listed in the script-name appendix.

`npc-library/archival-scripting-notes.md`

Archival Scripting Notes

This note records prior analysis of Classic Graal NPC scripting systems, performed outside this repository and archived on decommissioning of the analysis project (2026-10-28).

Source Archive

A full dump of extracted, catalogued, and documented NPC scripts plus architecture notes was packaged as **Graal Scripting.zip** (24 MB, ~3,900 files) and uploaded to Google Drive under `/[Grok/[Projects/`.

Direct link: (Drive link omitted from the document release)

The archive contains:

- **Grokd extract** — cleaned unique script bodies (1,520 unique from 6,275 placements), `CATALOG.tsv` with body-hashes, packs concatenated per system.
- **Phoenix documented copies** — same scripts with analysis headers (source path, classification, events hooked, functions, `toweapons` names, `client.*` / `save[]` usage, timeout rates, gotchas).
- **Ash architecture docs** — runtime model of the Classic client NPC engine (events, timeout scheduler, weapon-bar persistence, image ID bands), plus per-pack blueprints, flowcharts, algorithms, and pseudocode for larency HUD/inventory, Phoenix quests/farm, Meh weapons/smartbaddy, Ultimate NPC Pack minigames, and standalones.
- Original pack folders and assets (larency, ultimatenpcpack1, etc.).

Classification Summary (from Ash)

Across ~1,519 documented scripts:

Class	Approx. count	Typical home
quest / dialogue	516	Phoenix
level prop / trigger	390	UNP + Phoenix
farm / planting	136	Phoenix
player weapon / projectile	135	Meh + UNP
shop / merchant	111	UNP + larency
minigame	59	UNP + Phoenix
enemy / baddy AI	44	Meh + Phoenix
inventory / equipment UI	37	larency

Class	Approx. count	Typical home
HUD / stats overlay	20	larency

Dominant author tags: tricxta (HUD/inventory/shop), Ace2896 (weapons + smartbaddy), Agret (UNP houses/minigames), plus community.

Runtime Model Highlights (Classic client)

- Pure client-side NPCs. No separate server VM visible in the dump.
- Level NPC vs weapon NPC (`toweapons -Name`) vs putnpc children.
- Tick model is `timeout = N` (seconds) → `if (timeout)` wake; common rates 0.05 (AI/HUD), 0.1–0.2 (menus), 0.5–1.0 (idle).
- Persistence: `toweapons`, `save[n]`, `client.*` flags.
- Image ID collisions common in the ~300+ band used by HUDs and shops.

These notes are complementary to the existing `npc-library/` unique scripts and `gs1-reference-205.md`. They emphasize control flow, host effects, and real-world patterns useful for offline script engines.

Boundary

No raw decompiles or proprietary client code. All material is derived from public level dumps, observed script behavior, and open documentation. The archive is for reference only; it is not a playable server or complete asset set.

`npc-library/gs1-command-reference-205.md`

GS1 command reference (client 2.05 rev 2)

Date: 2026-10-21

What each GS1 command does in the 2.05 rev 2 client, read from the client's script compiler and command executor. Cross-checked against `npcprogramming.doc` (2000) and the `newfeatures*.txt` change logs shipped with Graal Reborn 2.22. Where the client and the docs disagree, the client wins and the row says so. **[inf]** marks a reading that is not fully pinned down.

This replaces the guesses in gs1-commands.md for 2.05. That file's pattern-letter list and its `warp` / `jump` rows were wrong; see "Corrections" at the end.

How the interpreter works

- Scripts are compiled once, on level load, into a flat list of statement nodes. Each node holds a command number, up to two string arguments, up to eight numeric expression arguments, a jump target, and a pre-resolved name index (item, baddy, colour, etc.).
- Command numbers are **600 + table index** in the GS1 numbering. The 2.05 table has **161** commands (index 0..160, opcodes 600..760). Earlier dumps listed 163 because they ran into the next table in memory, the direction names `up`, `left`, `down`, `right`.
- Each command has an argument signature. The compiler uses it to parse and to build the "usage" text in error messages:

Letter	Compiler's label	Accepts
`R`	number	any numeric expression
`M`, `S`	string	raw text, message codes expanded at run time
`F`, `Q`	filename	file name; offline, the compiler rejects a missing file unless it starts with `http://`. `Q` names a script file
`I`	itemname	one of 25 item names (table below)
`B`	baddyname	one of 10 baddy names
`C`	colorname	one of 20 colour names, or a number
`D`	direction	`up` `left` `down` `right` (0..3), or a numeric expression
`L`	carry objectname	one of 11 carry objects
`O`	objectname	an entry from the editor's predefined tile-objects list

- Every event runs the whole node list from the top. `if` evaluates its expression and jumps past its block when it is 0. `else`, `function` and the closing braces never reach the executor: the compiler turns them into four internal node kinds: assignment / expression statement, unconditional jump, function call, and end of function body.

- Loops (`for`, `while`) count passes and stop after **10,000**, as the doc says. Function calls push a return point and stop nesting at **10,000** deep. A `return` (or the end of a function body) with an empty call stack ends the script.

- Several commands **end the current script run** after they execute: `setlevel`, `setlevel2`, `destroy`, `sleep` (non-zero), `toinventory` (offline), `hide` on an NPC that is following the player, and `break` / `continue` outside a loop.

- String arguments are trimmed and message codes (`#c`, `#v(...)`, ...) are expanded before use, for almost every command.

- **Weapon NPCs act as the player.** When the running NPC is one of the player's weapons, commands that spawn things (bombs, explosions, horses, projectiles, `showimg` layers, `hitobjects`, `triggeraction`, `reflectarrow`) use the player's level and count the player as the source, not the NPC.

- **Online gates.** Several visual commands run only offline, or online when the server announced that it has an NPC server (server packet 44): `seteffect`, `setcoloreffect`, `setzoomeffect`, `changeimgcolors`, `changeimgzoom`, `addtiledef`, `addtiledef2`, `removetiledefs`, `replaceani`, `attachplayertoobj`. Online without an NPC server they do nothing.

- **Editor preview mode.** When the in-game editor builds a preview copy of a `showcharacter` NPC, it runs that NPC's `playerenters` in a restricted mode. Only `if`, `else`, `for`, `while`, `setgif`, `setgifpart`, `setimg`, `setimgpart`, `setmap`, `showcharacter`, `setcharprop`, `setcharani`, `setbackpal`, `setbacktile`, `setbacktile2`, `addtiledef`, `addtiledef2`, `setz`, `loadmap`, function calls, and `++`/`--` style assignments run. Everything else is skipped. **[inf: the trigger is the editor-copy path; exact UI action not traced]**

Name tables

- **Items** (`I`, index 0..24): greenrupee, bluerupee, redrupee, bombs, darts, heart, glove1, bow, bomb, shield, sword, fullheart, superbomb, battleaxe, goldensword, mirrorshield, glove2, lizardshield, lizardsword, goldrupee, fireball, fireblast, nukeshot, joltbomb, spinattack.
- **Baddies** (`B`, 0..9): graysoldier, bluesoldier, redsoldier, shootingsoldier, swampsoldier, frog, octopus, goldenwarrior, lizardon, dragon.
- **Colours** (`C`, 0..19): white, yellow, orange, pink, red, darkred, lightgreen, green, darkgreen, lightblue, blue, darkblue, brown, cynober, purple, darkpurple, lightgray, gray, black, transparent.
- **Carry objects** (`L`, 0..10): bush, sign, vase, stone, blackstone, bomb, hotbomb, superbomb, joltbomb, hotjoltbomb, none.
- **Directions** (`D`): up 0, left 1, down 2, right 3. Numeric values are reduced mod 4 (negative values too).

Online side effects

When connected, some commands also tell the server or nearby peers. TCP ids are PLI numbers from protocol/packet-ids.md; UDP ids are the peer-to-peer ids from protocol/udp-p2p.md.

Command	TCP (PLI)	UDP
`take`, `take2`, `removeitem`	13 ITEMDEL	20 item del
`putcomp`, `putnewcomp`	17 BADDYADD (instead of spawning locally)	
`putnpc`	21 PUTNPC (instead of creating locally)	
`destroy`	29 NPCWEAPONDEL if the NPC is also a weapon, then 22 NPCDEL	
`toweapons`	33 WEAPONADD	
`hitcompu`	16 BADDYHURT	
`hitplayer` (remote player)		21 remote hit, to the victim only
`putexplosion`, `putexplosion2`	27 EXPLOSION	14 explosion
`spyfire`	10 FIRESPY	13 fire spy
`hitobjects`	36 HITOBJECTS	25 hit objects
`triggeraction`	38 TRIGGERACTION	26 trigger action
`removebomb`		11 bomb del
`removehorse`, `takehorse`		17 horse del
`reflectarrow`		12 arrow (re-sent with the new direction)
`setplayerprop #c,...`	chat prop	22 chat

Plain TCP routing also applies: a level-local action goes over UDP only when every other player in the level has a UDP port, otherwise over both (see udp-p2p.md).

Commands

"Doc" means npcprogramming.doc or the newfeatures logs. Positions are in tiles unless noted.

Messages, signs, sound, files

Op	Command	Args	What 2.05 does
600	`say`	R	Shows sign *n* of the current level (index into the level's sign list). Skipped if the index is out of range or the player is already in the sign-reading state. Puts the player into the reading state.
711	`say2`	S	Same as `say`, but shows the given text directly instead of a level sign.
648	`message`	S	Sets the text drawn above the NPC. An empty string clears it. Also turns off `timershow`.
606	`play`	F	Plays a sound positioned at the NPC's centre. Online, a file that is not found locally is requested by its raw name. Doc: midis loop.
702	`playlooped`	F	Same, but the file is queued as a looped sound (internally the name gets a `+` prefix). Stops on level exit or `stopsound`.
703	`stopsound`	F	Stops that (looped) sound.
637	`stopmidi`	–	Stops the current midi.
739	`setmusicvolume`	R,R	Sets music volume, left and right (doc: 0..1).
635	`openurl`	M	Opens the URL in the system browser (doc: with a confirm dialog, no `http://` needed).
740	`openurl2`	M,R,R	Opens the URL in the client's own HTML window of the given width and height.
676	`showfile`	F	Opens the file with its associated program, like a double-click.
678	`setbackpal`	F	Copies the 256-colour palette of the given image into the tileset and into every loaded level's tile image, then repaints. Remembers the file so the same palette is not reloaded.
679	`setletters`	F	Replaces the sign font image (default `letters.png`) if the file exists.

Flags and variables

Op	Command	Args	What 2.05 does
604	`set`	M	Sets a flag (doc: `server.` prefix makes it server-wide).
607	`unset`	M	Clears a flag.
677	`setstring`	S,S	Sets string variable *name* = *value* (stored as the flag `name=value`). Empty value deletes it.

Op	Command	Args	What 2.05 does
666	`setarray`	R,R	Resizes the array variable named by the first argument to *n* elements. Ignored unless the first argument is a plain or `this.` variable.
742	`tokenize`	S	Splits the expanded string into the global token list on spaces and commas (quotes group).
743	`tokenize2`	S,S	Same, with extra delimiter characters added to the space/comma set.

Control flow

Op	Command	Args	What 2.05 does
605	`if`	–	Evaluates the condition; jumps past the block when it is 0.
634	`else`	–	Compiled away into a jump.
660	`for`	–	Loop; condition checked each pass; stops after 10,000 passes.
668	`while`	–	Same loop node as `for`.
667	`sleep`	R	Sets the NPC's timeout to the value and remembers the next statement; the next `timeout` event resumes there. `sleep 0` just clears the timer display. Ends the current run.
712	`break`	–	Jumps to the end of the enclosing loop. Outside a loop it ends the script.
713	`continue`	–	Jumps to the loop's next pass. Outside a loop it ends the script.
716	`function`	–	Compiled away. Calls to a named function become a call node (max depth 10,000).
717	`return`	–	Pops the call stack and resumes after the call. With an empty stack it ends the script.
710	`callnpc`	R,S	Runs NPC *index*'s script with the given event flag set. Doc: params after the event are passed as `#p(n)`.
714	`callweapon`	R,S	Moves weapon *index* to the player's centre, then runs its script with the given event flag set.
744	`triggeraction`	R,R,S,S	Fires *action* with *params* at the object on (x, y) locally, and sends it online. From a weapon, the sender is the player.

NPC appearance and behaviour

Op	Command	Args	What 2.05 does
608 / 719	`setgif` / `setimg`	F	Sets the NPC image, shows the NPC, and resets the visible part to the whole image.

Op	Command	Args	What 2.05 does
680 / 720	`setgifpart` / `setimgpart`	F,R,R,R,R	Same, then shows only (x, y, w, h) of the image in pixels. x, y are clamped to 0..16000 and w, h to 0..220, then trimmed to the image size.
759	`show`	–	Shows the NPC (re-applies its image).
686	`showlocal`	–	Same, but marked local (not synced).
760	`hide`	–	Hides the NPC (only if it is in the current level). If it was following the player, it stops following and the run ends.
687	`hidelocal`	–	Same, marked local.
688	`dontblock`	–	Player can walk through the NPC.
619	`dontblocklocal`	–	Same, local only.
689	`blockagain`	–	Blocks again and also resets the draw layer to normal (undoes `drawoverplayer`/`drawunderplayer`).
638	`blockagainlocal`	–	Same, local only.
620	`drawoverplayer`	–	Draw layer +1.
639	`drawunderplayer`	–	Draw layer −1.
729	`drawaslight`	–	Draw layer 9: above the day/night effect, below the status bar.
641 / 642	`canbecarried` / `cannotbecarried`	–	Player can / cannot lift the NPC.
644 / 645	`canbepushed` / `cannotbepushed`	–	Player can / cannot push it.
646 / 647	`canbepulled` / `cannotbepulled`	–	Player can / cannot pull it.
631	`timershow`	–	Draws the NPC's timeout counter over it. Cleared by `message` and `sleep 0`.
670	`timereverywhere`	–	This client gets the NPC's timeout events even when it is not the level leader.
725	`setshape`	R,R,R	Type 0 or 1 (1 = fixed rectangle), width and height in pixels. Other types are ignored.
751	`setshape2`	R,R,R	Makes the NPC block like a small level: width and height in tiles, plus a tile-type array (literal or array variable) of w×h entries. A short or missing array clears the shape.
726	`seteffect`	R,R,R,R	Global screen tint r, g, b, a (day/night). Online gate applies.
727	`setcoloreffect`	R,R,R,R	Per-NPC colour multiply r, g, b, a. Online gate applies.
728	`setzoomeffect`	R	Per-NPC zoom factor. Online gate applies.
745	`move`	R,R,R,R	Smoothly moves the NPC by dx, dy over *time* seconds with option bits. **[inf: option bits not decoded]**
653	`destroy`	–	Removes the NPC from the level (and from the weapon list), sends the deletes online, ends the run.
652	`putnpc`	Q,Q,R,R	Offline: creates a new NPC with that image and script file at (x, y) and runs its `playerenters`. Online: asks the server instead.

Character NPCs (`showcharacter`)

Op	Command	Args	What 2.05 does
673	`showcharacter`	–	Draws the NPC as a player character (once; later calls do nothing).
675	`setcharprop`	M,M	Sets a character property from a message code: `#1` sword image, `#2` shield image, `#3` head image, `#5` horse image (creates the horse if needed), `#8` body image, `#C0`..`#C5` colours (name or number), `#P1`..`#P5` attributes, `#c` chat text (not in preview mode), `#n` nickname. Other codes are ignored. The first word is the property code (not expanded as a message). Codes inside the **value** are expanded first; on script-running servers `#n` in that value is the triggering player's nickname, while property code `#n` writes the NPC's nickname — see `server-notes/setcharprop-hash-n.md`.
709	`setcharani`	S,S	Sets the character's gani and its parameter string.
733	`setchargender`	S	`male` sets male; anything else sets female.
684	`carryobject`	L	Character carries the object (0..10; `none` = 10). Only for `showcharacter` NPCs; switches to the carry gani.
685	`throwcarry`	–	Throws the carried object if the character is in a carry sprite; bombs become live bombs.
701	`takehorse`	R	Character mounts horse *index* (removes it from the level). Only for `showcharacter` NPCs.
633	`followplayer`	–	Offline only: NPC follows the player and stops blocking. Ignored online (matches the doc).

The player

Op	Command	Args	What 2.05 does
610	`setlevel`	M	Warps the player to that level at the current x, y. Ends the run.
705	`setlevel2`	M,R,R	Warps to that level at (x, y). Ends the run.
655	`sethead`	F	Player head image.
683	`setbody`	F	Player body image (skipped if it is the same file).
649	`setsword`	M,R	Sword image and power.
650	`setshield`	M,R	Shield image and power.

Op	Command	Args	What 2.05 does
663	`setbow`	S	Sets the bow image on the running NPC's character, not on the player. **[differs from doc, which says the player's bow]**
616	`setplayerdir`	D	Player direction, reduced mod 4. Accepts a name or an expression (doc says names only).
611..615	`setskincolor`, `setcoatcolor`, `setsleevecolor`, `setshoecolor`, `setbeltcolor`	C	Player colour slots 0..4.
674	`setplayerprop`	M,M	Same codes as `setcharprop` but on the player: `#1 #2 #3 #5 #8 #C0..#C5 #P1..#P5 #c`. `#c` sets the chat text with a 60-tick timeout and sends it online. `#7` is **not** handled (doc lists it).
708	`setani`	S,S	Player gani and parameter string.
749	`replaceani`	S,S	Replaces one of the 31 default gani names (walk, idle, sword, ...) with another gani file. Online gate applies.
732	`setgender`	S	`male` sets male; anything else sets female.
603	`hurt`	R	Hurts the player by half-hearts and knocks them away from the NPC's centre. Skipped when the player cannot be hurt right now (dead, in a protected state, or flagged unhittable on an NPC-server). 0 or negative heals (doc).
654	`freezeplayer`	R	Freezes the player for 0..30 seconds (clamped) and drops what they carry.
669	`hideplayer`	R	Hides the player for 0..30 seconds.
659	`hidesword`	R	Hides the sword for 0..30 seconds.
618	`takeplayercarry`	–	Removes what the player carries and resets the gani.
691	`takeplayerhorse`	–	Removes the player's horse.
621 / 622	`enableweapons` / `disableweapons`	–	Turns player weapons on / off.
737 / 738	`enabledefmovement` / `disabledefmovement`	–	Turns the built-in movement and controls on / off.
753 / 754	`enableselectweapons` / `disableselectweapons`	–	Allows / blocks opening the weapon selection.

Op	Command	Args	What 2.05 does
723	`showstats`	R	Bitmask of status-bar parts to show (doc: 1 ASD, 2 icons, 4 rupees, 8 bombs, ...).
750	`attachplayertoobj`	R,R	Type 0 only (NPC): attaches the player to the NPC with that id. Online gate applies.
752	`detachplayer`	–	Detaches the player if attached.
632	`toinventory`	M	Sets the flag and moves the NPC into the inventory. Offline it leaves the level and the run ends; online a copy goes to the inventory and the NPC stays. Only for NPCs of the current level.
656	`toweapons`	M	Adds a copy of the NPC as weapon *name* (replacing one with the same name, in the same slot), runs its `playerenters`, selects it, and sends it online. A `/` in the name keeps only the part after it.

Combat, items, level objects

Op	Command	Args	What 2.05 does
601	`lay`	I	Drops the item next to the NPC, to the right or below depending on where the player stands (doc).
699	`lay2`	I,R,R	Drops the item at (x, y).
602	`take`	I	Picks up the first item of that type lying within the NPC's radius (half its diagonal) and gives it to the NPC.
700	`take2`	R	Picks up item *index*.
694	`removeitem`	R	Deletes item *index*.
627	`putbomb`	R,R,R	Bomb of power 0..2 at (x, y).
628	`putexplosion`	R,R,R	Explosion of radius *r* at (x, y), with sound.
722	`putexplosion2`	R,R,R,R	Power, radius, x, y; power 3 is a jolt-bomb explosion.
629	`putleaps`	R,R,R	Leaps of type 0..5 at (x, y) (bush, swamp, stone, sign, ball, water).
640	`puthorse`	F,R,R	Horse with that image at (x, y).
692	`removebomb`	R	Deletes bomb *index*.
697	`explodebomb`	R	Detonates bomb *index* now.
693	`removearrow`	R	Deletes arrow *index* (local only, doc).

Op	Command	Args	What 2.05 does
698	`reflectarrow`	R	Reverses arrow *index*. Against a character NPC it bounces off the side it hits; otherwise both directions flip. The arrow's sprite turns to its new direction and the shooter becomes the player when called from a weapon.
695	`removeexplo`	R	Deletes explosion *index*.
696	`removehorse`	R	Deletes horse *index*.
623..626	`shootarrow`, `shootfireball`, `shootfireblast`, `shootnuke`	D	Fires that projectile from the NPC's edge in the given direction (reduced mod 4). Drawn on the NPC's layer, or the player's for weapons.
630	`shootball`	–	Fires a ball from the NPC's centre toward the player.
715	`spyfire`	R,R	Spawns a fire-spy effect just in front of the player (player position plus their facing direction) and sends it online. **[inf: role of the two arguments]**
662	`hitplayer`	R,R,R,R	index, half-hearts, fromx, fromy. Local player: hurt with knockback. Another player: UDP 21 to that player only (skipped if they are dead or protected). An NPC: treated like `hitnpc`.
706	`hitnpc`	R,R,R,R	NPC *index* gets the damage; its script gets `washit` style handling (no automatic push, doc).
741	`hitobjects`	R,R,R	Hits everything at (x, y) with *power*, like a sword, and sends it online.
617	`putcomp`	B,R,R	Offline: spawns that baddy at (x, y). Online: asks the server.
651	`putnewcomp`	B,R,R,M,R	Same, plus image and power.
661	`hitcompu`	R,R,R,R	index, power, fromx, fromy: damages and knocks back the baddy, and sends it online.
609	`removecompus`	–	Deletes every baddy (doc: causes `compusdied`).
643	`noplayerkilling`	–	Turns off player-vs-player damage in this level.

Board, map, images

Op	Command	Args	What 2.05 does
664	`updateboard`	R,R,R,R	Redraws the board rectangle (x, y, w, h) after `board[]` writes (doc: also syncs online).

Op	Command	Args	What 2.05 does
665	`putobject`	O,R,R	Offline only: stamps a predefined tile object on the 64×64 board at (x, y), clipped to the board, then redraws. Ignored online (doc: security).
672	`setmap`	F,F,R,R	Loads the big map image and its level-names file, sets the player's head-icon position, and recomputes "is on map" for every player. Both files must exist.
671	`setminimap`	F,F,R,R	Same for the minimap.
758	`loadmap`	F	Loads a gmap file; if the current level belongs to it (and the editor is closed), rebuilds the level view.
746	`addtiledef`	S,S,R	Use image *file* instead of `pics1.png` for levels starting with *prefix*. Online gate applies.
747	`addtiledef2`	S,S,R,R	Same, but pastes the image at (x, y) of the tileset. Online gate applies.
748	`removetiledefs`	S	Removes tile definitions for *prefix*. Online gate applies.
657	`showimg`	R,F,R,R	Image *index* at (x, y), on the NPC's layer +1 (player's for weapons). Re-using an index replaces it; the same file only moves. A name starting with `@` is a text image.
704	`changeimgpart`	R,R,R,R,R	Visible part of image *index*; x, y clamped to 0..16000, w, h to 0..220.
718	`changeimgvis`	R,R	Layer of image *index*: 0, 1, 2 = under / level with / over the NPC; 3 = light layer (like `drawaslight`); 4 = screen (over the status bar).
730	`changeimgcolors`	R,R,R,R,R	Colour multiply for image *index*. Online gate applies.
731	`changeimgzoom`	R,R	Zoom for image *index*. Online gate applies.
658	`hideimg`	R	Deletes image *index*.

Parsed but ignored by the 2.05 client

These are in the table, so scripts using them compile, but the executor has no handler. They do nothing in 2.05.

Op	Command	Args	Notes
636	`seturllevel`	M	Doc describes it as an offline URL warp; not implemented here.
681	`setfocus`	R,R	Later logs mention a camera focus at (x, y).
682	`resetfocus`	–	
690	`setbacktile`	R	Allowed in preview mode but still a no-op.
707	`canwarp2`	–	Warp permission is handled by the server in this era **[inf]**.
721	`drawovertrees`	–	

Op	Command	Args	Notes
724	`canwarp`	–	
734	`join`	S	Server-side class join in later versions **[inf]**.
735	`freezeplayer2`	–	
736	`unfreezeplayer`	–	
755	`setbacktile2`	R,R,R,R,R	
756	`setz`	R×8	
757	`updateterrain`	–	

Corrections to gs1-commands.md

- Pattern letters: `R` is a numeric expression, not the literal string "real". `D` is up 0, left 1, down 2, right 3 (or a number), not "0 south, 1 west, 2 east, 3 north".
- `warp` and `jump` are not GS1 commands in 2.05. Warping is `setlevel` / `setlevel2`, or assigning `playerx` / `playery`.
- `canwarp`, `canwarp2`, `join`, `setfocus`, `resetfocus`, `unfreezeplayer` compile but do nothing in 2.05.
- `spyfire` is a game effect with a network packet (PLI 10), not a debug hook.

`npc-library/gs1-commands.md`

GS1 command catalog (cleaned)

Date: 2026-10-14

Derived from the public command tables, the Reborn scripting guide, the 3,172 scraped scripts, and observed offline behavior. No decompile material.

> For what each command actually does in client 2.05, see gs1-command-reference-205.md. It corrects the pattern letters below (`R` is a number, `D` is up 0, left 1, down 2, right 3) and notes that `warp` / `jump` are not GS1 commands.

Control flow

- `if` / `else` — conditional. The nearby form is used for range checks against the player or other objects.
- `jump` — unconditional transfer to a label.
- `function` — named block. Early clients treat this as a no-op or simple call; later clients expand it.

The offline interpreter already executes arithmetic, comparison, and jumps. The missing piece is the command call itself.

Commands observed in the scraped set

Command	Typical effect	Notes
`hide`	Make the NPC invisible	Extremely common in `playertouchsme`
`show`	Make the NPC visible	Paired with hide
`setlevel`	Change the current level	Used by level-warp scripts
`warp`	Move the player to a new x,y	Local and cross-level
`hurt`	Apply damage to the player	Combat and hazard scripts

Command	Typical effect	Notes
`say`	Display a text box	Dialogue scripts
`set` / `unset`	Set or clear a flag	Triggers, doors, state
`setfocus`	Direct camera or attention	Rare in the scraped set
`resetfocus`	Clear focus	Rare
`spyfire`	Observation or debug hook	Present in public tables, rare in scripts
`loadmap`	Load a map overlay	Present in public tables
`canwarp` / `canwarp2`	Query whether a warp is allowed	Used before warp
`join`	Attach to another object	Rare
`freezeplayer` / `unfreezeplayer`	Lock or unlock player input	Used in cutscenes

The public command table also lists draw-layer commands (`setbacktile`, `drawovertrees`, `setz`, `updateterrain`) and a few others that do not appear in the 3,172 scraped scripts.

Pattern letters

When a command expands a template string, the letters are replaced as follows:

- B — baddy name
- C — color name
- D — direction (0 up, 1 left, 2 down, 3 right; corrected 2026-10-21)
- F / Q — filename
- I — item name
- L — carried object name
- M / S — arbitrary string
- O — object name
- R — numeric expression (corrected 2026-10-21; earlier read as the literal “real”)

These letters appear in both the public tables and in a small number of the scraped scripts.

Offline wiring priority

CmdCall is currently a no-op. The highest-value first wires, based on frequency in the scraped set, are:

1. `hide` / `show` — appears in the majority of `playertouchsme` scripts.
2. `warp` / `setlevel` — appears in transport and level-change scripts.
3. `hurt` — appears in combat and hazard scripts.
4. `say` — appears in dialogue scripts.
5. `set` / `unset` — appears in trigger and door scripts.

Wiring the first two is enough for an offline session to respond to player touch by hiding an NPC or warping the player. That is the smallest change that makes a level feel interactive instead of static.

Event frequency (reminder)

From the 3,172 scripts:

- `playerenters` — 2,268
- `playertouchsme` — 1,986
- `timeout` — 1,456

- `playerendsreading` — 1,123
- `compusdied` — 1,122
- `playertouchsother` — 1,074
- `wasshot` — 728

Most of these events call one of the commands listed above.

Update 2026-10-21

- The full 2.05 command table (161 entries, op = 600 + index, 600–760) and the function/variable table (ids 100–825) are in `gs1-reference-205.md`, together with built-in event flags, constants, and differences from 2.22.
- 2.05 interprets a parse tree built from script **text**. It does not run OpenGS1-style bytecode. The `Call`/`CmdCall` gap in Seihai is therefore a command-dispatch problem: map each command name to a host function, using the table's argument signatures.
- `show` (759) and `hide` (760) are the last two entries of the 2.05 table. Earlier dumps that stopped at 758 missed them; dumps that listed 163 entries ran on into the direction-name table (`up`, `left`, …). **[2.05]**
- `setfocus`/`resetfocus`, `spyfire`, `loadmap`, `canwarp`/`canwarp2`, `join`, `freezeplayer`, and `unfreezeplayer` are all present in 2.05. `freezeplayer` takes a time argument.

`npc-library/gs1-reference-205.md`

GS1 reference — client 2.05 rev 2

Date: 2026-10-21

Sources: the 2.05 command and expression tables (names and argument signatures only; no code); Graal Reborn 2.22 `commands.rtf` and `npcprogramming.doc` (user's Drive); OpenGS1 (MIT) for comparison. Tags: **[2.05]** read from the 2.05 tables · **[doc]** 2.22 docs · **[inf]**.

Execution model [2.05]

- 2.05 does **not** compile GS1 to bytecode. NPC and weapon scripts arrive as source text (`NPCPROP_SCRIPT`), are parsed into a tree, and are interpreted recursively.
- Command ids are `600 + table index`. `if` is 605. A condition counts as true when it evaluates to ≥ 1.0.
- Expressions resolve through a second table of functions and variables with numeric ids (100–825, below) and a small operator table.
- OpenGS1's stack bytecode (Push/Assign/Jmp/…) is an independent reimplementation. Its opcode set is not something 2.05 shares. Keep Seihai's VM, but load scripts from text.
- Command names that end in a space in the table (`say`, `set`, `setlevel`, `setimg`, `move`, …) require an argument. The others are matched as whole words. **[inf]**

Argument signature letters

One letter per argument in the table's signature column:

Letter	Meaning	Tag
R	numeric expression	[2.05][inf]
S	string expression (`#s`, quoted text, string vars)	[inf]
M	message / free text up to `;` (flag names, level names)	[inf]

Letter	Meaning	Tag
F / Q	filename	[inf]
I	item name (`greenrupee` … `spinattack`)	[doc]
B	baddy name (`graysoldier` … `dragon`)	[doc]
C	color name (`white` … `transparent`)	[doc]
D	direction: `up`/`left`/`down`/`right` or 0–3	[doc]
L	carry object (`bush`, `sign`, `vase`, `stone`, `blackstone`, `bomb`, …)	[doc]
O	tile object (from `tileobjects/`)	[inf]

This replaces the "R = literal `real`" reading in `gs1-commands.md`.

Commands (161) [2.05]

Op	Command	Args	Op	Command	Args	Op	Command	Args
600	`say`	R	601	`lay`	I	602	`take`	I
603	`hurt`	R	604	`set`	M	605	`if`	—
606	`play`	F	607	`unset`	M	608	`setgif`†	F
609	`removecompus`	—	610	`setlevel`	M	611	`setskincolor`	C
612	`setcoatcolor`	C	613	`setsleevecolor`	C	614	`setshoecolor`	C
615	`setbeltcolor`	C	616	`setplayerdir`	D	617	`putcomp`	BRR
618	`takeplayercarry`	—	619	`dontblocklocal`	—	620	`drawoverplayer`	—
621	`enableweapons`	—	622	`disableweapons`	—	623	`shootarrow`	D
624	`shootfireball`	D	625	`shootfireblast`	D	626	`shootnuke`	D
627	`putbomb`	RRR	628	`putexplosion`	RRR	629	`putleaps`	RRR
630	`shootball`	—	631	`timershow`	—	632	`toinventory`	S
633	`followplayer`	—	634	`else`	—	635	`openurl`	M
636	`seturllevel`	M	637	`stopmidi`	—	638	`blockagainlocal`	—
639	`drawunderplayer`	—	640	`puthorse`	MRR	641	`canbecarried`	—
642	`cannotbecarried`	—	643	`noplayerkilling`	—	644	`canbepushed`	—
645	`cannotbepushed`	—	646	`canbepulled`	—	647	`cannotbepulled`	—
648	`message`	S	649	`setsword`	MR	650	`setshield`	MR
651	`putnewcomp`	BRRMR	652	`putnpc`	QQRR	653	`destroy`	—
654	`freezeplayer`	R	655	`sethead`	F	656	`toweapons`	M
657	`showimg`	RMRR	658	`hideimg`	R	659	`hidesword`	R
660	`for`	—	661	`hitcompu`	RRRR	662	`hitplayer`	RRRR
663	`setbow`†	S	664	`updateboard`	RRRR	665	`putobject`	ORR
666	`setarray`	RR	667	`sleep`	R	668	`while`	—
669	`hideplayer`	R	670	`timereverywhere`	—	671	`setminimap`	FFRR
672	`setmap`	FFRR	673	`showcharacter`	—	674	`setplayerprop`	MM
675	`setcharprop`	MM	676	`showfile`	F	677	`setstring`	SS
678	`setbackpal`	F	679	`setletters`	F	680	`setgifpart`†	FRRRR
681	`setfocus`	RR	682	`resetfocus`	—	683	`setbody`	F
684	`carryobject`	L	685	`throwcarry`	—	686	`showlocal`	—
687	`hidelocal`	—	688	`dontblock`	—	689	`blockagain`	—
690	`setbacktile`†	R	691	`takeplayerhorse`	—	692	`removebomb`	R
693	`removearrow`	R	694	`removeitem`	R	695	`removeexplo`	R
696	`removehorse`	R	697	`explodebomb`	R	698	`reflectarrow`	R
699	`lay2`	IRR	700	`take2`	R	701	`takehorse`	R
702	`playlooped`	F	703	`stopsound`	F	704	`changeimgpart`	RRRRR

Op	Command	Args	Op	Command	Args	Op	Command	Args
705	setlevel2	MRR	706	hitnpc	RRRR	707	canwarp2†	—
708	setani	SS	709	setcharani	SS	710	callnpc	RS
711	say2	S	712	break	—	713	continue	—
714	callweapon	RS	715	spyfire	RR	716	function	—
717	return	—	718	changeimgvis	RR	719	setimg	F
720	setimgpart	FRRRR	721	drawovertrees†	—	722	putexplosion2	RRRR
723	showstats	R	724	canwarp†	—	725	setshape	RRR
726	seteffect	RRRR	727	setcoloreffect	RRRR	728	setzoomeffect	R
729	drawaslight	—	730	changeimgcolors	RRRRR	731	changeimgzoom	RR
732	setgender	S	733	setchargender	S	734	join†	S
735	freezeplayer2†	—	736	unfreezeplayer	—	737	enabledefmovement	—
738	disabledefmovement	—	739	setmusicvolume	RR	740	openurl2	MRR
741	hitobjects	RRR	742	tokenize	S	743	tokenize2	SS
744	triggeraction	RRSS	745	move	RRRR	746	addtiledef	SSR
747	addtiledef2	SSRR	748	removetiledefs	S	749	replaceani	MM
750	attachplayertoobj	RR	751	setshape2	RRR	752	detachplayer	—
753	enableselectweapons	—	754	disableselectweapons	—	755	setbacktile2†	RRRRR
756	setz†	RRRRRRRR	757	updateterrain	—	758	loadmap	F
759	show	—	760	hide	—			

† = not in the 2.22 `commands.rtf`: either removed or renamed by 2.22 (`setgif` → `setimg`, `setgifpart` → `setimgpart`), or undocumented (`setz`, `drawovertrees`, `setbacktile*`, `canwarp*`, `join`, `freezeplayer2`, `setbow`).

Table size: 161 commands (600–760). `show` (759) and `hide` (760), the two most common commands in the scraped library, are the last two entries; an earlier dump stopped before them. The two records after 760 belong to the direction-name table, not to the command table. Which commands compile but have no executor case is in `gs1-command-reference-205.md`. Commands documented for 2.22 but absent in 2.05 include `addstring`/`insertstring`/`replacestring`/`removestring`/`deletestring`, `putnpc2`, `showtext`/`showtext2`, `showpoly`, `showani`, `play2`, `setshootparams`, `shoot`, `enablefeatures`, `seteffectmode`, `changeimgmode`, and the mouse/keypressed family.

Functions and variables [2.05]

Idrange	Group	Names
100–102	board/conversion	`board[R]`, `strtofloat(S)`, `timevar`
200–218	objectcountsandglobals	`compuscount`, `playerscount`, `bombscount`, `arrowscount`, `itemscount`, `exploscount`, `horsescount`, `npcscount`, `signscount`, `selectedweapon`, `s`
400–455	math, tests, stringtests	`random(R,R)`, `sin(R)`, `cos(R)`, `arctan(R)`, `int(R)`, `abs(R)`, `arraylen(R)`, `testcompu(R,R)`, `testplayer(R,R)`, `testbomb(R,R)`, `testitem(R,R)` `onmapy(S)`, `vecx(R)`, `vecy(R)`, `getz(R,R)`
600–606	compus[i].*	7 members; the 2.22 docs list the same count: x, y, type, dir, headdir, power, mode [doc]

Id range	Group	Names
640–643	bombs[i].*	4 members; the 2.22 docs list the same count: x, y, power, time [doc]
660–666	arrows[i].*	7 members; the 2.22 docs list the same count: x, y, dx, dy, dir, type, from [doc]
680–683	items[i].*	4 members; the 2.22 docs list the same count: x, y, type, time [doc]
700–704	explos[i].*	5 members; the 2.22 docs list the same count: x, y, power, time, dir [doc]
720–726	horses[i].*	7 members; the 2.22 docs list the same count: x, y, dir, bushes, bombs, bombpower, type [doc]
740–758	this.*/npcs[i].*	`x`, `y`, `width`, `height`, `rupees`, `darts`, `bombs`, `hearts`, `timeout`, `glovepower`, `swordpower`, `shieldpower`, `hurtdx`, `hurtdy`, `id`, `save[2]`, `dir`, `sprite`
780–781	signs[i].*	2 members; the 2.22 docs list the same count: x, y [doc]

Id range	Group	Names
800–819	players[i].*	`playerx`, `playery`, `playerrupees`, `playerdarts`, `playerbombs`, `playerhearts`, `playerfullhearts`, `playerdir`, `playerglovepower`, `playerbomb
820–825	players[i].*	`playerhurtdy`, `playerfreezetime`, `playerhurtpower`, `playerattachid`, `playerattachtype`, `playerz` (the same ids reached through `players[i]`

Member names for `compus[i]`, `bombs[i]`, and similar objects are not in the expression table. Each group's member count matches the 2.22 doc list exactly. The names in the table are the doc names in doc order, and the mapping of names to ids is **[inf]** until the member parser is traced.

Built-in flags and events [2.05]

These are flags tested with `if (name)`, not functions. The engine sets them before running a script.

Event flags: `created`, `playerenters`, `timeout`, `playertouchsme`, `playertouchsother`, `playerendsreading`, `playerchats`, `playersays(…)`/`playersays2(…)` (functions), `wasshot`, `wasshooted`, `washit`, `wasthrown`, `waspelt`, `peltwith…`, `exploded`, `compusdied`, `compsdead`, `playerdies`, `playerhurt`, `playerhurted`, `playerlaysitem`, `weaponfired`, `action…` (any `triggeraction` name), `shotbyplayer`, `shotbybaddy`.

State flags: `visible`, `carrying`, `carries…`, `carriesnpc`, `weaponsenabled`, `followsplayer`, `isweapon`, `playeronhorse`, `canspin`, `playerswimming`, `issparringzone`, `sparringzone`, `nopkzone`, `isleader`, `playerismale`, `playerisfemale`, `playerreading`, `isonmap`, `lighteffectsenabled`, `playerattached`, `firedonhorse`.

`main` sits in the same string cluster. It is probably the weapon's main event or entry. **[inf]**

Absent in 2.05 but documented for 2.22: `actionleftmouse`/`actionrightmouse`/`actiondoublemouse`, `actionprojectile`, `keypressed`, `mousedown`, `playeronline`, `playerpaused`.

Constants [doc]

- **Items** (`lay`, `take`, chest item index): 0 greenrupee, 1 bluerupee, 2 redrupee, 3 bombs, 4 darts, 5 heart, 6 glove1, 7 bow, 8 bomb, 9 shield, 10 sword, 11 fullheart, 12 superbomb, 13 battleaxe, 14 goldensword, 15 mirrorshield, 16 glove2, 17 lizardshield, 18 lizardsword, 19 goldrupee, 20 fireball, 21 fireblast, 22 nukeshot, 23 joltbomb, 24 spinattack. GServer's `LevelItemType` agrees. **[srv]**
- **Baddies:** 0 graysoldier, 1 bluesoldier, 2 redsoldier, 3 shootingsoldier, 4 swampsoldier, 5 frog, 6 octopus, 7 goldenwarrior, 8 lizardon, 9 dragon. 2.05 also has `baddyhare.gif`, so a hare type exists. Its index is open.
- **Baddy modes:** 0 walking, 1 looking, 2 hunting, 3 hurt, 4 bumped, 5 dying, 6 shooting (swamp), 7 jumping (frog), 8 shooting (spider), 9 dead.
- **Directions:** 0 up, 1 left, 2 down, 3 right.
- **Player sprites:** 0 idle; 1–8 walk; 9–13 sword; 14–18 push; 19–22 pull; 23 lift; 24 carry idle; 25–32 carry walk; 33 shoot; 34–36 ride; 37 sit; 38 sleep; 39 hurt; 40 dead (39 and 40 only on `showcharacter` NPCs).
- **Tile types** (`tiletype(x,y)`): 0 nonblocking, 2 hurting underground, 3 chair, 4 bed top, 5 bed bottom, 6 swamp, 7 lava swamp, 8 near water, 9 near lava, 11 water, 12 lava, 20 throw-through, 21 jumping stone, 22 blocking.
- **Colors:** white, yellow, orange, pink, red, darkred, lightgreen, green, darkgreen, lightblue, blue, darkblue, brown, cynober, purple, darkpurple, lightgray, gray, black, transparent. The same list is used for `PLPROP_COLORS` indexes. **[srv]**
- **`move` options:** cachetype 0–2, +4 block check, +8 event when done, +16 apply direction.
- **`seteffectmode`** (2.22): 0 light, 1 transparency, 2 holes.

Operators and syntax [doc]

`=`, `+=`, `-=`, `*=`, `/=`, `++`, `--`; arithmetic `+ - * / %`, `^` power; comparison `== != < > <= >=`; logic `&& || !`; `in` for ranges and sets (`x in |0,64|`, `2 in {1,2,3}`); `this.` for NPC-local variables; `#` string codes (`#s(var)`, `#v(expr)`, …, 2.x). Control flow: `if`/`else`, `for`, `while`, `break`, `continue`, `function name() {}` + `name();`, `return`, `sleep`, `timeout = n`.

The full `#` string-code set for 2.05 is not yet extracted. **Open.**

`npc-library/README.md`

NPC library

3,172 unique scripts (the scrape reported 3,187, but 15 records were lost to truncated source pages; see `partial_pages` in the `npcs.json` meta), broken into a hierarchy so no single file is large.

Structure

```
genre / topic / book-NN / chapter-NN / page.json
```

- **Genre** — combat, transport, object, dialogue, spawn, interaction, misc
- **Topic** — targets, hazards, baddies, weapons, level-warps, local-warps, signs, doors, chests, blocks, speech, timers, visibility, flags, triggers, other
- **Book** — up to 100 scripts
- **Chapter** — up to 25 scripts
- **Page** — one `page.json` per chapter

Counts

Genre	Scripts	Topics
combat	1,400	targets (728), baddies (452), hazards (192), weapons (28)
interaction	602	triggers (411), visibility (180), flags (11)
dialogue	373	speech (373)
spawn	328	timers (328)
misc	257	other (257)
object	168	doors (108), blocks (35), signs (15), chests (10)
transport	59	local-warps (32), level-warps (27)

Index

`index.json` lists every genre and topic with script and book counts.

Reference

`reference/` holds the small NPC material from shanalysis: - `npcControl-NPC.txt` — GServer-v2 control NPC header - `forcesay.txt` — phoenix execscript

Tile patterns

`tile-patterns.md` is unchanged.

Flat file

`npcs.json.bak` is the original 1 MB flat dump, kept for reference. Use the hierarchy.

GS1 references

- `gs1-commands.md`: cleaned catalog and offline wiring priority.
- `gs1-reference-205.md`: the full 2.05 command and function tables, built-in flags, and constants.

`npc-library/tile-patterns.md`

Tile orientation patterns

Scanned 1420 GR-V1 .graal levels.

Most common horizontal neighbors (left → right)

- `6966` → `2028` × 4552
- `6c61` → `7965` × 3440
- `706c` → `6179` × 3338
- `6179` → `6572` × 3310
- `a769` → `6620` × 2348
- `7da7` → `6966` × 2172
- `2870` → `6c61` × 2127

- `6620` → `2870` × 1885
- `2028` → `706c` × 1757
- `2920` → `7b20` × 1635
- `7468` → `6973` × 1588
- `207b` → `207d` × 1538
- `6869` → `732e` × 1508
- `207d` → `a769` × 1430
- `7b20` → `7da7` × 1360
- `3ba7` → `2020` × 1039
- `6e77` → `616c` × 994
- `616c` → `6c28` × 985
- `2e67` → `6966` × 957
- `2069` → `6620` × 946

Most common vertical neighbors (above → below)

- `706c` → `2920` × 263
- `2920` → `6620` × 255
- `7965` → `7761` × 253
- `6966` → `706c` × 250
- `6966` → `6572` × 243
- `6572` → `a769` × 238
- `a769` → `7368` × 236
- `2e35` → `2020` × 235
- `7965` → `7c7c` × 225
- `7274` → `2077` × 225
- `6f75` → `6173` × 224
- `6c61` → `6420` × 223
- `6c61` → `207b` × 207
- `2920` → `2028` × 203
- `2028` → `7574` × 202
- `6179` → `7b20` × 201
- `6620` → `6f74` × 200
- `6966` → `656f` × 200
- `7da7` → `696d` × 198
- `6620` → `6c6f` × 198

`server-notes/setcharprop-hash-n.md`

`#n` in `setcharprop` (server brief A3.6)

Date: 2026-10-31

Question: when an original server runs `setcharprop` and the **value** contains `#n`, does that expand to the **NPC's** nickname or the **player's**?

Answer

On servers that run GS1 themselves, `#n` inside a `setcharprop` **value** expands to the **action player's** nickname (the player who triggered the script), not the NPC's. Optional `#n(id)` picks another player by server id. Tag: **[srv]**

The property **code** `#n` (first argument) is separate: it writes the expanded value into the NPC's own nickname field. Tag: **[srv]**

Evidence

1. GServer13 (1999) cannot settle the value question by lab

The Linux `gserver13` binary from the GServer 1.31 source tree (dated 1999-12-22) and its matching C++ sources have player and NPC **props** (including an NPC `nickname` field) but **no** GS1 command table and **no** `setcharprop` / `showcharacter` strings. NPC scripts in that era were carried to clients and run there; the 1999 server did not interpret `setcharprop`. Tag: **[bin]** (binary and source inventory under Drive `GServer 1.31`; not committed).

So a Wine loopback of GServer13 + a 2.05 client does not exercise server-side `#n` expansion. That path only appears once the server takes over NPC scripting.

2. Server-side GS1 (GServersource archive)

The later server NPC engine (`npcs.cpp` in the Drive `GServersource` tree) does the following. Tag: **[srv]** (plain behaviour from that source; no code quoted).

1. Message-code expansion (`GetRealMessage`) treats `#n` as the nickname of `actionplayer` (default), or of `GetPlayer(id)` when written `#n(id)`.
2. `setcharprop` always runs the **second** argument through that expander before applying it.
3. When the **first** argument is `#n`, the expanded text is stored as both the NPC object's nickname and the `showcharacter` avatar's nickname, and marks the NPC nickname prop dirty for send.

So `setcharprop #c,Hi #n` becomes chat text containing the **player's** nick, and `setcharprop #n,Guard` sets the **NPC's** nick to `Guard` (after expanding any codes inside `Guard`).

GServer 1.34's Windows binary lists `showcharacter` but not `setcharprop`, so it is not a useful lab host for this command either. Tag: **[bin]**

3. Client 2.05 (for the online client-run path)

`npc-library/gs1-command-reference-205.md`: string arguments are trimmed and message codes are expanded before use; `setcharprop`'s first argument names the property (`#n` = nickname on the character NPC). Tag: **[2.05]**

`setplayerprop` on 2.05 does **not** handle property code `#n` (only `#1 #2 #3 #5 #8 #C0..#C5 #P1..#P5 #c`). Nickname changes for a character figure go through `setcharprop`. Tag: **[2.05]**

A loopback capture that proves the server-side expansion on a live script-running server (GServer-v2 lab or similar) with 2.05 rev 2 is still open. Tag: **[lab]** needed.

What Seihai / forks should keep

- Expand `#n` in `setcharprop` **values** as the triggering player's nickname (same as other message text).
- Do not substitute the NPC's current nickname for bare `#n` in that value.
- Keep property code `#n` as "set this NPC's nickname".

Still open

- **[lab]** Run a level NPC under a script-running lab server with `setcharprop #c,seen:#n;` (and a distinct NPC nick via `setcharprop #n,Npc`) and confirm the chat/NPC props on the wire match the player nick. Preferred host: lab GServer-v2 on loopback; client 2.05 rev 2 under Wine. GServer13 is the wrong host for this command.
- Whether any pre-GServersource Windows server already ran `setcharprop` (none found in the downloadable GServer 1.19 / 1.31 / 1.34 packages).